① $\sum_{n\ge1}\frac{\sqrt{n^3}}{\sqrt{n^7+5n^q+7}-n^3\sqrt{n}}$ $q\in(0,7)$

$$\sum_{n\ge1}\frac{\sqrt{n^3}\left(\sqrt{n^7+5n^q+7}+n^3\sqrt{n}\right)}{\cancel{n^7}+5n^q+7-\cancel{n^7}}=\sum_{n\ge1}\frac{n^5\left(\sqrt{1+5n^{q-10}+\frac{7}{n^{10}}}+1\right)}{5n^q+7}$$

$v_n=\frac{1}{n^{q-5}}$

$$\lim_{n\to\infty}\frac{u_n}{v_n}=\lim_{n\to\infty}\frac{n^5\left(\sqrt{1+5n^{q-10}+\frac{7}{n^{10}}}+1\right)}{5n^q+7}\cdot n^{q-5}=$$

$$=\lim_{n\to\infty}\frac{\cancel{n^5}\left(\sqrt{1+5n^{q-10}+\frac{7}{n^{10}}}+1\right)}{\cancel{n^q}\left(5+\frac{7}{n^q}\right)}\cdot\cancel{n^{q-5}}=\frac{1+1}{5}=\frac{2}{5}\in(0,+\infty)$$

$\xRightarrow{Cr.\ III}\sum u_n\sim\sum v_n$ (serii convergente)

pt $q-5>1\Rightarrow q>6\Rightarrow\sum x_n$ convergentă

pt $q-5\le1\Rightarrow q\le6\Rightarrow\sum x_n$ divergentă

② $\sum_{n\ge1}\frac{(an)^n}{n!}$, $a>0$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{[a(n+1)]^{n+1}}{(n+1)!}\cdot\frac{n!}{(an)^n}=\lim_{n\to\infty}\frac{a(n+1)^{n+1}}{(n+1)\cdot n^n}$$

$$=\lim_{n\to\infty}\frac{a(n+1)^n}{n^n}=\lim_{n\to\infty}a\cdot\left(\frac{n+1}{n}\right)^n=a\cdot\lim_{n\to\infty}\left(\frac{n+1}{n}\right)^n=a\cdot e$$

Dacă $a\cdot e<1\Rightarrow a<\frac{1}{e}\xRightarrow[Rap]{Crit.}\sum x_n$ convergent

Dacă $a\cdot e>1\Rightarrow a>\frac{1}{e}\xRightarrow[Rap]{Crit}\sum x_n$ divergent

Dacă $a\cdot e=1\Rightarrow a=\frac{1}{e}\Rightarrow\sum_{n\ge1}\frac{1}{e^n}\cdot\frac{n^n}{n!}=\sum_{n\ge1}\frac{1}{\cancel{e^n}}\cdot\frac{\cancel{n^n}}{\left(\frac{n}{\cancel{e}}\right)^n\cdot\sqrt{2\pi n}}=$

$=\sum_{n\ge1}\frac{1}{\sqrt{2\pi n}}\xrightarrow[I_4]{Crit.}\sum_{n\ge1}\frac{1}{\sqrt{2\pi n}}\sim\sum\frac{1}{\sqrt{n}}\Rightarrow\sum x_n$ divergentă

Formula lui Stirling

$$\boxed{n!\simeq\left(\frac{n}{e}\right)^n\cdot\sqrt{2\pi n}}$$

③ $\sum_{n\ge1}\frac{a(a+1)\dots(a+n-1)}{n^2\cdot n!}$ $a>0$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{\cancel{a(a+1)\dots(a+n-1)}(a+n)}{(n+1)^2\cdot\underset{n+1}{\cancel{(n+1)!}}}\cdot\frac{n^2\cdot\cancel{n!}}{\cancel{a(a+1)\dots(a+n-1)}}=$$

$$=\lim_{n\to\infty}\frac{(a+n)\cdot n^2}{(n+1)^3}=\lim_{n\to\infty}\frac{\cancel{n^3}\left(1+\frac{a}{n}\right)}{\cancel{n^3}\left(1+\frac{1}{n}\right)^3}=1$$

$$\lim_{n\to\infty}n\cdot\left(\frac{x_n}{x_{n+1}}-1\right)=\lim_{n\to\infty}n\left[\frac{(n+1)^3}{n^2(a+n)}-1\right]=\lim_{n\to\infty}\frac{(n+1)^3-n^2(a+n)}{n(a+n)}$$

$$=\lim_{n\to\infty}\frac{\cancel{n^3}+3n^2+3n+1-an^2-\cancel{n^3}}{n(a+n)}=\lim_{n\to\infty}\frac{(3-a)n^2+3n+1}{n^2\left(1+\frac{a}{n}\right)}=3-a$$

Dacă $3-a>1\Rightarrow a<2\Rightarrow\sum_{n\ge1}x_n$ convergentă

Dacă $3-a<1\Rightarrow a>2\Rightarrow\sum_{n\ge1}x_n$ divergentă

Dacă $a=2\Rightarrow\sum_{n\ge1}\frac{2\cdot3\cdot\ldots\cdot(n+1)}{n^2\cdot n!}=\sum_{n\ge1}\frac{n+1}{n^2}\sim\sum_{n\ge1}\frac{1}{n}\xRightarrow[I_4]{Crit}$

$\Rightarrow\sum_{n\ge1}x_n$ diverg.

④ $\sum_{n\ge1}\frac{2\cdot5\cdot8\cdot\ldots\cdot(3n-1)}{1\cdot5\cdot9\cdot\ldots\cdot(4n-3)}\cdot\frac{1}{2^n(3n+1)}$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{\cancel{2\cdot5\ldots(3n-1)}(3n+2)}{\cancel{1\cdot5\ldots(4n-3)}(4n+1)}\cdot\frac{1}{2^{\cancel{n}+1}(3n+4)}\cdot\frac{\cancel{1\cdot5\ldots(4n-3)}}{\cancel{2\cdot5\ldots(3n-1)}}\cdot\cancel{2^n}(3n+1)$$

$$=\lim_{n\to\infty}\frac{3n+2}{4n+1}\cdot\frac{3n+1}{2(3n+4)}=\frac{3\cdot\cancel{3}}{4\cdot2\cdot\cancel{3}}=\frac{3}{8}<1\Rightarrow\sum_{n\ge1}x_n\text{ converg.}$$

⑤ $\sum_{n\ge1}\frac{(2n)!}{n^n}$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(2n+2)!}{(n+1)^{n+1}}\cdot\frac{n^n}{(2n)!}=\lim_{n\to\infty}\frac{(2n+1)(2n+2)\cdot n^n}{(n+1)^{n+1}}$$

$$=\lim_{n\to\infty}\frac{(2n+1)\overset{2}{\cancel{(2n+2)}}}{\cancel{n+1}}\cdot\left(\frac{n}{n+1}\right)^n=\lim_{n\to\infty}2(2n+1)\cdot\left(\frac{n}{n+1}\right)^n=2\cdot\infty\cdot e=\infty$$

$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\infty\Rightarrow\sum_{n\ge1}^{\infty}x_n$ e divergentă

⑥ $\sum_{n\ge1}\frac{(2n-1)!!}{(2n)!!}\cdot\frac{a^n}{2n+1}$, $a>0$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{\overset{2n+1}{\cancel{(2n+1)!!}}}{\underset{2n+2}{\cancel{(2n+2)!!}}}\cdot\frac{a^{\cancel{n}+1}}{2n+3}\cdot\frac{\cancel{(2n)!!}}{\cancel{(2n-1)!!}}\cdot\frac{2n+1}{\cancel{a^n}}$$

$$=\lim_{n\to\infty}\frac{2n+1}{2n+2}\cdot\frac{2n+1}{2n+3}\cdot a=a$$

$a>1\Rightarrow\sum_{n\ge1}x_n$ divergentă

$a<1\Rightarrow\sum_{n\ge1}x_n$ convergentă

$a=1\Rightarrow\sum_{n\ge1}x_n=\sum_{n\ge1}\frac{(2n-1)!!}{(2n)!!}\cdot\frac{1}{2n+1}$

$$\frac{1}{2\sqrt{n}}<\frac{(2n-1)!!}{(2n)!!}<\frac{1}{\sqrt{2n+1}}\Big|\cdot\frac{1}{2n+1}\Rightarrow\frac{1}{2(2n+1)}<x_n<\frac{1}{(2n+1)^{\frac{3}{2}}}\not\Rightarrow$$

$\Rightarrow\sum\frac{1}{(2n+1)\sqrt{2n+1}}\sim\sum\frac{1}{n^{\frac{3}{2}}}$

convergentă pt. că $\alpha=\frac{3}{2}>1$ (Serie Riemann)

$\xRightarrow[I]{Crit}\sum_{n=1}^{\infty}x_n$ convergentă

⑦ $\sum_{n\ge1}\frac{1\cdot5\cdot9\cdot\ldots\cdot(4n-3)}{3\cdot7\cdot11\cdot\ldots\cdot(4n-1)}\cdot\alpha^n$, $\alpha>0$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{\cancel{1\cdot5\ldots(4n-3)}(4n+1)}{\cancel{3\cdot7\ldots(4n-1)}\cdot(4n+3)}\cdot\alpha^{n+1}\cdot\frac{\cancel{3\cdot7\ldots(4n-1)}}{\cancel{1\cdot5\ldots(4n-3)}}\cdot\frac{1}{\alpha^n}=$$

$$=\lim_{n\to\infty}\frac{4n+1}{4n+3}\cdot\alpha=\alpha$$

Dc. $\alpha>1\Rightarrow\sum_{n=1}^{\infty}x_n$ divergentă

Dc. $\alpha<1\Rightarrow\sum_{n=1}^{\infty}x_n$ convergentă

Dc. $\alpha=1\Rightarrow\sum_{n\ge1}\frac{1\cdot5\ldots(4n-3)}{3\cdot7\ldots(4n-1)}\xRightarrow[R-D]{Crit.}\lim_{n\to\infty}n\left(\frac{x_n}{x_{n+1}}-1\right)=\lim_{n\to\infty}n\cdot\left(\frac{4n+3}{4n+1}-1\right)$

$$=\lim_{n\to\infty}n\cdot\frac{\cancel{4n}+3-\cancel{4n}-1}{4n+1}=\lim_{n\to\infty}\frac{2n}{4n+1}=\frac{1}{2}<1\Rightarrow\sum_{n=1}^{\infty}x_n\text{ converg.}$$

⑧ $\sum_{n\ge1}\cos\frac{1}{\sqrt{1}}\cdot\cos\frac{1}{\sqrt{2}}\cdots\cos\frac{1}{\sqrt{n}}$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{\cancel{\cos\frac{1}{\sqrt{1}}\cdots\cos\frac{1}{\sqrt{n}}}\cdot\cos\frac{1}{\sqrt{n+1}}}{\cancel{\cos\frac{1}{\sqrt{1}}\cdots\cos\frac{1}{\sqrt{n}}}}=\lim_{n\to\infty}\cos\frac{1}{\sqrt{n+1}}=\cos0=1$$

$$\lim_{n\to\infty}n\left(\frac{1}{\cos\frac{1}{\sqrt{n+1}}}-1\right)=\lim_{n\to\infty}n\left(\frac{1-\cos\frac{1}{\sqrt{n+1}}}{\underbrace{\cos\frac{1}{\sqrt{n+1}}}_{\to1}}\right)=\lim_{n\to\infty}n\cdot2\sin^2\frac{1}{2\sqrt{n+1}}=$$

$$=\lim_{n\to\infty}2n\cdot\underbrace{\frac{\sin^2\frac{1}{2\sqrt{n+1}}}{\left(\frac{1}{2\sqrt{n+1}}\right)^2}}_{\to1}\left(\frac{1}{2\sqrt{n+1}}\right)^2=\lim_{n\to\infty}\cancel{2}n\cdot\frac{1}{\underset{2}{\cancel{4}}(n+1)}=\frac{1}{2}\Rightarrow$$

$\Rightarrow\sum_{n\ge1}x_n$ divergentă

⑨ $\sum_{n\ge1}\left(\frac{1}{n}-\ln\frac{n+1}{n}\right)$

$$\ln\frac{n+1}{n}>\frac{1}{n+1}\Rightarrow\frac{1}{n}-\ln\left(\frac{n+1}{n}\right)<\frac{1}{n}-\frac{1}{n+1}=\frac{1}{n^2+n}$$

$\sum_{n\ge1}\frac{1}{n^2+n}\sim\sum_{n\ge1}\frac{1}{n^2}$ ↳ convergentă $\mapsto\sum_{n\ge1}x_n$ convergentă

⑩ $\sum_{n\ge1}\frac{1}{4^n\cos^2\frac{x}{2^n}}$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{4^n\cdot\cos^2\frac{x}{2^n}}{4^{n+1}\cos^2\frac{x}{2^{n+1}}}=\frac{1}{4}\lim_{n\to\infty}\frac{\cos^2\frac{x}{2^n}}{\cos^2\frac{x}{2^{n+1}}}=$$

$=\frac{1}{4}\cdot1\Rightarrow\sum_{n\ge1}x_n$ conv

⑪ $\sum_{n\ge1}\frac{n^p\ln n}{n!}$, $p\in\mathbb{R}$

$$\lim_{n\to\infty}\frac{x_{n+1}}{x_n}=\lim_{n\to\infty}\frac{(n+1)^p\ln(n+1)}{\underset{n+1}{\cancel{(n+1)!}}}\cdot\frac{\cancel{n!}}{n^p\ln n}=\lim_{n\to\infty}\underbrace{\left(\frac{n+1}{n}\right)^p}_{1}\cdot\overbrace{\frac{\ln(n+1)}{\ln n}}^{1}\cdot\overbrace{\frac{1}{n+1}}^{0}$$

$=1\cdot1\cdot0=0<1\Rightarrow\sum_{n=1}^{\infty}x_n$ converg.

⑫ $\sum_{n\ge1}\frac{a^n\ln n}{n}$, $a>0$

$$\lim_{n\to\infty}\frac{a^{\cancel{n}+1}\ln(n+1)}{n+1}\cdot\frac{n}{\cancel{a^n}\ln n}=\lim_{n\to\infty}a\cdot\underset{\downarrow\atop1}{\frac{n}{n+1}}\cdot\underset{\downarrow\atop1}{\frac{\ln(n+1)}{\ln n}}=a$$

$\xRightarrow[Rap.]{Crit.}$ $a>1\Rightarrow\sum_{n\ge1}^{\infty}x_n$ divergentă

$a<1\Rightarrow\sum_{n\ge1}^{\infty}x_n$ convergentă

$a=1\Rightarrow\sum_{n\ge1}^{\infty}x_n=\sum_{n\ge1}^{\infty}\frac{\ln n}{n}$

$\frac{\ln n}{n}>\frac{1}{n}\Rightarrow\sum_{n\ge1}^{\infty}\frac{\ln n}{n}>\sum_{n\ge1}^{\infty}\frac{1}{n}\xrightarrow[I]{Crit}$

$\sum_{n\ge1}^{\infty}\frac{1}{n}$ divergentă

$\Rightarrow\sum_{n\ge1}x_n$ divergentă

⑬ $\sum_{n\ge1}\left[\sqrt{(n+1)(n+a)}-n\right]^n$

$\xRightarrow[Rad.]{Crit.}\lim_{n\to\infty}\sqrt[n]{x_n}=\lim_{n\to\infty}\left[\sqrt{(n+1)(n+a)}-n\right]=\lim_{n\to\infty}\frac{\cancel{n^2}+(a+1)n+a-\cancel{n^2}}{\sqrt{(n+1)(n+a)}+n}=$

$$=\lim_{n\to\infty}\frac{(a+1)\cdot n+a}{n\left[\sqrt{\left(1+\frac{1}{n}\right)\left(1+\frac{a}{n}\right)}+1\right]}=\frac{a+1}{2}$$

pt. $\frac{a+1}{2}>1\Rightarrow a>1\Rightarrow\sum x_n$ diverg.

pt. $\frac{a+1}{2}<1\Rightarrow a<1\Rightarrow\sum x_n$ converg.

pt. $\frac{a+1}{2}=1\Rightarrow a=1\Rightarrow\sum_{n\ge1}\left[\sqrt{(n+1)^2}-n\right]^n=\sum_{n\ge1}1$ diverg.

Crit. Necesar pt. că $\lim_{n\to\infty}1=1\ne0$

⑭ Suma seriei $\sum_{n\ge1}\left(1+\text{tg}\frac{1}{n}\cdot\text{tg}\frac{1}{n+1}\right)\cdot\text{tg}\frac{1}{n(n+1)}$

$$\text{tg}\frac{1}{n(n+1)}=\text{tg}\frac{(n+1)-n}{n(n+1)}=\text{tg}\left(\frac{1}{n}-\frac{1}{n+1}\right)=\frac{\text{tg}\frac{1}{n}-\text{tg}\frac{1}{n+1}}{1+\text{tg}\frac{1}{n}\cdot\text{tg}\frac{1}{n+1}}$$

$$x_n=\left(1+\text{tg}\frac{1}{n}\cdot\text{tg}\frac{1}{n+1}\right)\cdot\frac{\text{tg}\frac{1}{n}-\text{tg}\frac{1}{n+1}}{1+\text{tg}\frac{1}{n}\cdot\text{tg}\frac{1}{n+1}}=\text{tg}\frac{1}{n}-\text{tg}\frac{1}{n+1}$$

$$S_n=\sum_{k=1}^{n}\left(\text{tg}\frac{1}{k}-\text{tg}\frac{1}{k+1}\right)=\text{tg}1-\text{tg}\frac{1}{n+1}$$

$\lim_{n\to\infty}S_n=\text{tg}1-\text{tg}0=\text{tg}1=\frac{\pi}{4}\Rightarrow\sum_{n=1}^{\infty}x_n$ convergent

pt. că S_n converg.

⑮ $\sum_{n\ge1}n\left(2^{\frac{3n}{n^2+1}}-3^{\frac{n^2}{n^3+1}}\right)$

$$\lim_{n\to\infty}x_n=\lim_{n\to\infty}n\left(2^{\frac{3n}{n^2+1}}-3^{\frac{n^2}{n^3+1}}\right)=$$

$$=\lim_{n\to\infty}n\left[2^{\frac{3n}{n^2+1}}-1-\left(3^{\frac{n^2}{n^3+1}}-1\right)\right]=$$

$$=\lim_{n\to\infty}n\cdot\left(\frac{2^{\frac{3n}{n^2+1}}-1}{\frac{3n}{n^2+1}}\cdot\frac{3n}{n^2+1}-\frac{3^{\frac{n^2}{n^3+1}}-1}{\frac{n^2}{n^3+1}}\cdot\frac{n^2}{n^3+1}\right)$$

$$=\lim_{n\to\infty}\left(\underbrace{\frac{2^{\frac{3n}{n^2+1}}-1}{\frac{3n}{n^2+1}}}_{\ln2}\cdot\overbrace{\frac{3n^2}{n^2+1}}^{3}-\underbrace{\frac{3^{\frac{n^2}{n^3+1}}-1}{\frac{n^2}{n^3+1}}}_{\ln3}\cdot\overbrace{\frac{n^3}{n^3+1}}^{1}\right)$$

$=3\ln2-\ln3\ne0\xRightarrow[necesar]{Crit}\sum_{n\ge1}x_n$ diverg.